

Types of Numbers, HCF and LCM (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	2 700 014	1	B1
1(b)(i)	1 or 64	1	B1
1(b)(ii)	6 or 10 or 21	1	B1
1(b)(iii)	$\frac{5}{3}$	1	B1 (accept $1\frac{2}{3}$)
1(c)	$2^3 \times 3 \times 5$	2	M1 for correct division by prime numbers
1(d)(i)	10	1	B1
1(d)(ii)	200	2	M1 for $2^3 \times 5^2$ or lists of multiples
2(a)	42 700 000 004	1	B1
2(b)(i)	27	1	B1
2(b)(ii)	28	1	B1 for 28 (since $T_7 = \frac{7 \times 8}{2} = 28$)
2(b)(iii)	π	1	B1 (note: $\sqrt{16} = 4$ is rational)
3(a)	Thirty million forty thousand and eight	1	B1 (allow minor spelling slips but numbers must be correct)
3(b)	15	2	M1 for listing factors of each or prime factorisations ($45 = 3^2 \times 5$ and $105 = 3 \times 5 \times 7$)
4(a)	Four million two thousand and fifty	1	B1
4(b)(i)	$\sqrt{9}$	1	B1 (since $\sqrt{9} = 3$)
4(b)(ii)	$\sqrt{10}$ or π	1	B1
4(c)	$2^3 \times 3^2$	2	M1 for correct prime factors 2, 2, 2, 3, 3 seen
4(d)(i)	12	1	B1 for $2^2 \times 3$ or 12
4(d)(ii)	2520	2	M1 for $2^3 \times 3^2 \times 5 \times 7$

Part	Answer	Marks	Partial Marks / Guidance
5(a)	$2^3 \times 3^2$	2	M1 for prime factor tree or continuous division showing correct prime factors (e.g. 2, 2, 2, 3, 3) or 8×9 written with one prime factorised correctly
5(b)	$\frac{1}{72}$	1	B1
6(a)	6	2	M1 for identifying common factors 2×3 or writing out $A = 60$ and $B = 126$ and listing common factors
6(b)	1260	2	M1 for $2^2 \times 3^2 \times 5 \times 7$ or for a clear listing method of multiples
7(a)	420	2	M1 for prime factorisations $42 = 2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$, or for listing multiples
7(b)	19:37	2	M1 for converting 420 seconds to 7 minutes A1 for adding 7 minutes correctly to 19:30
8(a)	3 040 007	1	B1
8(b)(i)	True	1	B1
8(b)(ii)	False	1	B1
8(b)(iii)	False	1	B1 (accept True if defined in non-strict context, but 0607 standard $\mathbb{N} = \{1, 2, 3, \dots\}$)
8(c)	$2 \times 3^2 \times 5$	2	M1 for prime factors 2, 3, 5 found
8(d)(i)	15	1	B1
8(d)(ii)	90	2	M1 for list of multiples or prime factor decomposition